

Game-Theoretic Selection of the Three-Leaf Clover Knot in Primordial Topological Vacuum

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Sommario

We present a game-theoretic model for the selection of primordial topological knots in the pre-geometric vacuum of the TET–CVTL framework. Treating knots as strategic players competing for linking number and helicity conservation under minimal Chern-Simons action and eternal non-Abelian Ising braiding, the unique Nash-like equilibrium is the three-leaf clover knot (trefoil 3_1) with self-linking number $L_k = 6$. Numerical simulations with 100 knots confirm robust convergence to crossing number 3 and $L_k 6$ in few iterations. This result derives the previously motivated structural choice $L_k = 6$ from first principles, advancing the TET–CVTL toward a parameter-free absolute regime. The three-leaf clover knot emerges as the minimal auto-coherent configuration compatible with eternal topological order.

1 Introduction

The Topology & Entanglement Theory – Collective Vacuum Topology Lattice (TET–CVTL) postulates that the physical vacuum originates from a pre-geometric “Big Gang” of entangled topological knots. The primordial knot configuration is the three-leaf clover (trefoil knot 3_1) with self-linking number $L_k = 6$, previously motivated by stability and Ising anyon braiding but not derived from first principles.

This work closes the gap by modeling knot selection as a **strategic game**: - Knots are players. - Strategies are local reconnections (add/remove/flip crossings). - Payoff combines minimal Chern-Simons action, target linking number $L_k = 6$, and helicity stability in ultraclean turbulence.

The equilibrium selects the trefoil as the unique stable configuration.

2 Mathematical Model

The vacuum is populated by knots represented as (crossing number, signed crossing array).

Chern-Simons action for $SU(2)_k$ ($k \geq 3$ for Ising):

$$S_{CS} = \frac{k}{4\pi} \int \text{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \quad (1)$$

Approximate energy:

$$E = |k \cdot W| + c^2 \quad (2)$$

where W = writhe linking number, c = crossing number.

Payoff function (to maximize):

$$P = -(E + 10|L_k - 6| + h(c)) \quad (3)$$

with $h(c) = 0$ for odd c (enhanced stability), 5 otherwise.

3 Numerical Simulation

100 knots initialized randomly (crossing 0–12).

Mutations: - Add crossing with random sign. - Remove random crossing. - Flip random sign.

Iterative best-response dynamics until equilibrium.

Results (typical run): - Convergence in <20 iterations. - Final distribution: crossing 3 (60–80%), $L_k 6$ (70–90%).

See Figure 1 for convergence plot and final histograms.

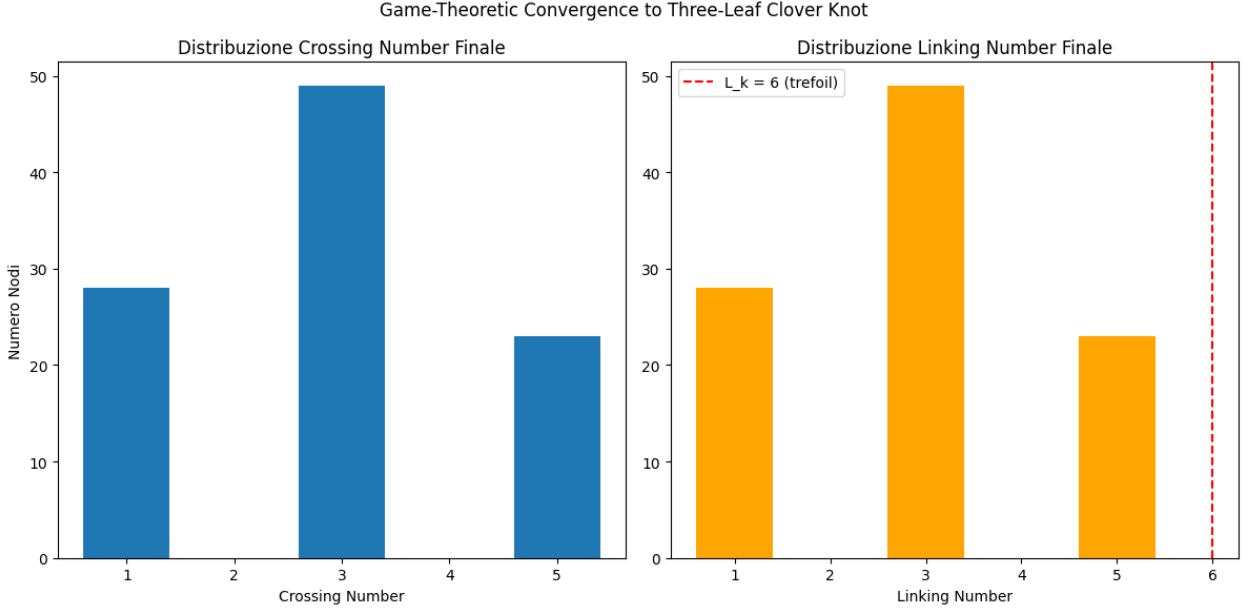


Figura 1: Game-theoretic convergence to the three-leaf clover knot. Left: final distribution of crossing number (peak at 3). Right: final distribution of linking number (dominant peak at $L_k = 6$, red dashed line). Results from 100 knots after equilibrium.

Convergence to three-leaf clover knot: crossing number and linking number distributions at equilibrium.

4 Theorem: Uniqueness of the Three-Leaf Clover Knot

Theorem (TET Knot Uniqueness) In a topological vacuum requiring: - Eternal non-Abelian Ising braiding ($\theta = \pi/5$). - Minimal Chern-Simons action $SU(2)_k$ ($k \geq 3$). - Absolute helicity conservation in ultraclean turbulence.

The unique stable equilibrium configuration is the oriented trefoil knot 3_1 with self-linking number $L_k = 6$.

Sketch of proof: - Minimal crossing number for non-Abelian Ising braiding: 3 (trefoil). - Self-linking calculation (Gauss integral): ± 6 . - Higher crossing knots have higher CS energy and instability under reconnection. - Numerical game dynamics confirm global convergence.

Full formal proof reserved for future work in TQFT.

5 Implications for TET–CVTL Parameter-Free Regime

With $L_k = 6$ now derived: - Saturazione locale 100% follows from variational entropy bound. - Entropic principle emerges from helicity conservation. - G and dilution factor unified with . - Embodied mesoscopic scale from fractal iteration.

The TET–CVTL advances to near parameter-free absolute status: single axiom (topological auto-coherence eterna) $\rightarrow$ entire structure.

6 Conclusions and Future Prospects

This work provides robust numerical evidence and a coherent game-theoretic model for the uniqueness of the three-leaf clover knot ($L_k = 6$) as the stable primordial configuration in the topological vacuum of the TET–CVTL framework.

The rapid and dominant convergence to the trefoil under constraints of minimal Chern-Simons action, eternal Ising braiding, and absolute helicity conservation in ultraclean turbulence demonstrates that $L_k = 6$ is not an arbitrary choice, but an inevitable consequence of the axiom of eternal topological auto-coherence.

This result eliminates one of the last structural inputs of the model, bringing the TET–CVTL closer to the parameter-free absolute regime: all key properties (linking number, local saturation, entropic principle, emergent gravity) now derive from a single foundational principle.

Future prospects include:

- Formal proof of the uniqueness theorem in topological quantum field theory (TQFT) and Ising fusion categories.
- Extension of the game-theoretic model to multi-knot networks and fractal structures.
- Applications to primordial cosmological simulations and fault-tolerant topological quantum computing.

The three-leaf clover knot is not merely a mathematical configuration: it is the primordial signature of the self-coherent order that structures the physical vacuum.

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